JAHANVI GAJERA

Roll No.: M25CSA012
Master of Technology in Artificial Intelligence
Computer Science Engineering
Indian Institute of Technology Jodhpur

+91-9327706381
jahanvigajera0106@gmail.com
m25csa012@iitj.ac.in
Github
LinkedIn

Education

Degree/Certificate	Institution/Board	CGPA/Percentage	Year
M.Tech in Artificial Intelligence	Indian Institute of Technology, Jodhpur	8.13	2025–Present
B.Tech in CSE	Rashtriya Raksha University, Gandhinagar	8.37	2019–2023
Class XII	GSHSEB	81.69%	2019
Class X	GSHSEB	91.83%	2017

Projects

•Research Area: Bias Mitigation in federated learning *IIT Jodhpur*
(Privacy-preserving federated learning models for Monkeypox detection under biased client data) June-2026 – present

- Designing a federated aggregation pipeline using class-wise entropy filtering and DBSCAN clustering to address non-IID client bias, consistently outperforming baseline methods in ongoing trials.
- Outperformed baseline federated aggregation across 4 vision architectures (DenseNet-121, ResNet-18, ResNet-50, MobileNetV3), boosting accuracy by up to 15.18% and achieving a peak accuracy of 91.42% and F1-score of 91.09% with our combined method.
- **Tech Stack:** PyTorch, Scikit-learn, NumPy, Python, CUDA | **Methods:** DBSCAN clustering, Information Entropy, Non-IID Federated Learning | **Architectures:** ViT, MobileNetV3, ResNet-18/50, DenseNet-121

•Sentiment Analysis
Github

- Designed a production-grade MLOps pipeline on Google Cloud Run to fine-tune and serve a DistilBERT sentiment analysis model, utilizing DVC and MLflow for experiment tracking and model registry rollbacks. The architecture exposes FastAPI and Streamlit services via a cost-efficient GitHub Actions CI pipeline and implements live statistical drift detection (PSI/JS/KS) to monitor incoming inference data continuously.
- **Tools:** MLflow, DVC, GitHub Actions, Docker, Google Cloud Run, FastAPI, Streamlit, Hugging Face, PyTorch.

•Distributed Semantic Search
Github

- Built a highly scalable, meaning-driven search engine that understands the context behind user queries, processing nearly a million Stack Exchange records using Apache Spark. To ensure both speed and reliability by combining GPU-accelerated AI models with FAISS vector indexing to deliver highly accurate search results in under 30 milliseconds.
- **Tools:** Sentence Transformers, Apache Parquet, Flask, Pandas, Python.

Key Courses Taken

- Machine Learning
- Deep Learning
- Computer Vision
- Natural Language Understanding
- ML-DL-Ops

Technical Skills

- **Programming Languages:** Python, Java, C, C++
- **Tools & OS:** Git, Linux, PyTorch, NumPy, Pandas, VS Code, Docker
- **Machine Learning:** PCA, Linear Regression, Logistic Regression, Support Vector Machines | **Computer Vision:** CNN, Vision Transformers | **Natural Language Processing (NLP):** Word Embeddings, RNN, LSTM, GRU, Transformers, BERT, GPT

Internship

Java Intern at Knovos India Private Limited *Jan-April 2023*

- Developed a distributed language detection engine using Java and Apache Hadoop MapReduce as part of an eDiscovery Document Assessment System, capable of identifying **26 languages** and their relative percentages in multilingual documents.

Academic Achievements

GATE CSE 2025: Qualified with a competitive score and secured admission to a top IIT.